REBECCA CHERY

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EDUCATION

University of California - Berkeley **Berkeley, CA**
BS in Electrical Engineering and Computer Science *Graduation Date: August 2023*
Relevant Coursework:
Data Structures • Operating Systems & System Programming • Efficient Algorithms & Intractable Problems • Computer Security • Great Ideas in Computer Architecture • Database Systems • Embedded Systems • Designing Information Devices & Systems

TECHNICAL SKILLS

Languages: Python • JavaScript • TypeScript • C/C++ • SQL • Java • Go • Rust
ML & Data: PyTorch • TensorFlow • Hugging Face • pandas • NumPy • Weights & Biases • CUDA • OpenMP
Frontend & Backend: React • FastAPI • Node.js • Docker • Kubernetes • CI/CD (Jenkins) • GCP • AWS • Terraform • REST APIs

WORK EXPERIENCE

Founder – PolyFlow (Pre-seed AI Speech Startup) | New York, NY | **polyflow.xyz** **Apr 2026–Present**
- Founded an AI speech platform for under-resourced languages, growing a 1,000+ person waitlist with 42% organic referrals and validating demand through 8+ user interviews and early pilots with learners, educators, and diaspora organizations
- Architected a production-ready SaaS platform using FastAPI, Next.js, PostgreSQL, and Redis, implementing authentication, relational data models, API caching, and modular backend services powering AI pronunciation coaching and multilingual learning tools
- Engineered multilingual NLP and speech pipelines that transformed raw multilingual corpora into 26,800+ curated training examples and 60+ hours of personally recorded Haitian Creole speech, enabling LLM/ASR fine-tuning and AI-powered pronunciation assessment

Software Engineer (Contract) – TEKsystems & Independent Clients | New York, NY **Aug 2023–Present**
- Led 15+ client engagements from architecture to production, building full-stack applications and AI tools with React, TypeScript, Python, Docker, and GCP while partnering directly with stakeholders to deliver business-critical solutions
- Engineered automated cloud data pipelines that accelerated large-scale data processing by 5×, reducing infrastructure complexity and improving delivery velocity

Software Engineer Intern – WarnerMedia | Remote **Jun–Aug 2021**
- Upgraded internal Play Portal deployment software using AWS Services, Node.js with TypeScript, and Python automation, enabling live branch builds and continuous deployments for Hadron (HBOMax) and reducing deployment request times by over 50 minutes
- Implemented a Slack App integration in Python to deliver real-time deployment updates, improving visibility and collaboration across developer teams

Software Development Engineer Intern – Amazon | Seattle **Jun–Aug 2019 & 2020**
- Implemented a send-to-curator playlist feature in Riddler using Ruby on Rails, React.js, and MySQL, driving a 20% increase in playlist submissions within the first month
- Designed and optimized REST APIs and built a scalable Music Station Feedback Service API in Java and MySQL to streamline playlist publishing, analyze performance metrics, and reduce latency for internal tools
- Collaborated with UX designers and back-end engineers in agile sprints to analyze user feedback, prioritize feature development, and deliver impactful updates for Amazon Music

TECHNICAL PROJECTS

Semantic Audio Map | *React Flow, Next.js, TypeScript, Tailwind CSS, OpenAI Whisper, GPT-4o-mini* **GitHub**
- Built a voice-to-graph pipeline using Whisper and GPT structured outputs to transcribe audio and extract key ideas into a structured node-edge JSON mind map
- Implemented an interactive editable canvas for semantic representation, with node linking, relabeling, and persistence to support iterative human-in-the-loop refinement

Audio Memory Map | *Python, Streamlit, FastAPI, PostgreSQL, pgvector, AWS S3, Folium* **GitHub**
- Built a spatial audio memory app with Streamlit and FastAPI for recording, transcribing, and geotagging voice memories on an interactive map
- Created a unified storage layer supporting local JSON/disk storage and cloud deployment with AWS S3 and PostgreSQL/pgvector
- Implemented REST APIs for memory upload, retrieval, audio streaming, and health checks in a modular full-stack architecture

Play-by-Ear Music Instrument Tutor | *Python, Streamlit, Librosa, NumPy, SciPy* **GitHub**
- Developed an audio analysis pipeline using Librosa pYIN for pitch tracking, sequence alignment for melody matching, and note-level scoring to evaluate performance accuracy
- Implemented rhythm and intonation feedback logic to detect timing drift, pitch deviations, and substitution errors across recorded or live instrument input